TAIM: Mechanism and System

I. What is TAIM

TAIM is a decentralized compute finance protocol. It creates a native, circulating currency for the core factor of production in the AI era—computing power.

It is not a compute rental platform, not a GPU futures exchange, not another "AI + blockchain" Frankenstein. It is an entirely new currency creation system: it takes the Tokens that the AI industry generates every day and is about to discard, locks them as collateral through smart contracts with over-collateralization, and mints a currency—TAIM—that is floored by real physical demand and capped by physical scarcity.

In one sentence: TAIM turns computing power itself into money.

II. Participants in the System

The TAIM system has three types of roles, each bearing distinct functions, together forming a self-circulating economy.

2.1 Minters

Minters are the currency creators of TAIM. They possess idle AI computing resources—this can be the spare GPU clusters of large data centers, mining rigs scattered across the globe, or anyone holding unused AI model invocation quotas.

Minters take their computing output over a future period—standardized, verifiable AI model Tokens—lock them as collateral in TAIM's smart contract pools, and then mint a corresponding amount of TAIM currency.

Key Design: Over-collateralization. Minters must lock Tokens exceeding the value of the TAIM they mint. This ensures that every unit of TAIM in the system has sufficient real computing power as value backing, preventing malicious minting and systemic inflation.

2.2 Users

Users are the real demand side of TAIM. They are AI developers, small startup teams, independent researchers—anyone who needs to invoke large-model computing power but is suffocated by the high pricing of centralized APIs.

Users exchange TAIM for real computing power in the market. When the market price of TAIM falls below the official API retail price, users can directly buy TAIM on the secondary market, then use TAIM to redeem the Tokens locked by minters and run their own AI models.

They provide real demand for the system. This is the essential difference between TAIM and other purely speculative cryptocurrencies: it has a value floor constituted by real use value.

2.3 Speculators

Speculators are the liquidity providers of TAIM. They trade TAIM on the secondary market, chasing profits from price fluctuations.

Speculators do not care about AI models themselves, nor about the technical parameters of Tokens. What they care about is TAIM's exchange rate movements, market sentiment, and changes in the supply-demand relationship of computing power.

They inject liquidity into the system. Without speculators, it would be difficult for users to obtain TAIM quickly at a fair price when needed. They are the market makers of this market, the lubricant that enables the price discovery mechanism to function.

III. Dual-Engine Stabilization Mechanism

This is the most core design of the TAIM system. It ensures that the price of TAIM will neither rise indefinitely nor fall to zero.

3.1 Ceiling Cap: Physical Scarcity

The price of TAIM will not rise indefinitely, because the total global computing power is finite.

What happens when TAIM's price soars far beyond the actual cost of computing power? Countless minters, attracted by profit, will connect their previously idle computing power to the TAIM network and mint more TAIM to sell. The influx of new computing power increases the supply of TAIM and suppresses the price.

But the total supply of computing power is not infinite. Global GPU production capacity is constrained by wafer fab capacity, electricity supply is constrained by power plant installations, and data center expansion is constrained by land and cooling systems. These physical limitations constitute the natural ceiling of TAIM's price.

This is using the laws of physics to cap a currency.

3.2 Floor Support: Real Demand

The price of TAIM will also not fall to zero, because when it falls low enough, real users will enter the market.

When the market price of TAIM falls below the official API retail price, any rational AI developer will stop buying services directly from centralized platforms and turn to buying TAIM to redeem computing power. They are not speculators; they do not care about coin price fluctuations. They care about only one thing: running their AI models at the lowest possible cost.

The entry of these users constitutes the value iron floor of TAIM. They buy TAIM not to hoard, but to consume on the spot. This consumption behavior permanently reduces the circulating supply of TAIM, automatically pushing the price back up until it returns to equilibrium.

3.3 Alternating Work of the Dual Engines

These two mechanisms are not simple "ceilings" and "floors," but a pair of pistons working in alternation:

- When speculators flood in and prices surge → Users are deterred by high prices, trading volume shrinks → Minters, attracted by profit, expand supply → Prices fall back.
- When speculators flee and prices plummet → Users, attracted by low prices, enter the market to sweep up and consume TAIM → Circulating supply decreases → Prices rebound, paving the way for the return of speculators.

The departure of speculators summons the entry of users. The consumption by users creates scarcity for the return of speculators. These two forces—greed and rationality—no longer prey upon each other, but become engines working in alternation within a self-balancing system.

This is nearly inexhaustible liquidity.

IV. Cross-Model Floating Exchange Rate

4.1 Why a Floating Exchange Rate is Needed

A Token from a GPT-5-level model and a Token from a small open-source model clearly should not be of equal value. The computing cost behind them, the quality of the intelligence output, and their market scarcity are vastly different. If Tokens from all models were forced to exchange for TAIM at a fixed rate, the entire system would be instantly pierced by arbitrageurs.

4.2 Pricing Mechanism

TAIM adopts a floating exchange rate based on actual transaction volume.

Every minter can set an initial exchange rate for the computing power Tokens they have staked as collateral. Users can choose to accept this price, or look elsewhere for a better quote. Through countless free-bidding transactions, the market forms a real-time, transparent average transaction price for the Token output of each model.

This is not just an exchange mechanism; it is a cross-model value measurement system. For the first time, the capabilities of different AI models globally will have a relative value that is traded out by real money in a free market and measurable in the same unit (TAIM).

If the Token exchange rate of a model continues to decline, the market is voting with its feet: its capability or demand is falling. If a new model's exchange rate surges after launch, it indicates market recognition of its disruptive potential.

TAIM will become the decentralized evaluation system for the value of global AI models.

V. Essential Differences from Physical Delivery Futures

Historically, all financial innovations attempting to anchor to the physical world ultimately slid into one of two abysses: the credit collapse of physical delivery, or the infinite speculation of cash settlement.

TAIM has no delivery date. It is not a promise for the future, but a direct right of withdrawal against Tokens already in the pool. It compresses the forward risk of "futures" into the mathematical certainty of "instant ownership confirmation."

VI. Conclusion

TAIM is not a perfect system. It is merely the only solution under the current capital dilemma of the AI industry that can logically take effect quickly. It is only responsible for getting capital running, lifting sentiment, and making the market less tense. But where it runs to is still an unknown.